Appendix E – ROI Calculation Methodology

This appendix documents the financial logic and scenario controls used to generate Automat-IA's ROI, payback, and benefit projections. All calculations are performed under the **Conservative Scenario Framework**, ensuring every figure in this report is realistic, risk-adjusted, and auditable.

E.1 Core Financial Formulas

Return on Investment (ROI)

ROI (%)=Net BenefitTotal Costs×100\text{ROI (\%)} = \frac{\text{Net Benefit}}{\text{Total Costs}} \times 100ROI (%)=Total CostsNet Benefit×100

Components

Net Benefit=Total Annual Benefits−Total Annual Costs\text{Net Benefit} = \text{Total Annual Benefits} - \text{Total Annual Costs}Net Benefit=Total Annual Benefits−Total Annual Costs Total Costs=Implementation+Software+Training+Change Management\text{Total Costs} = \text{Implementation} + \text{Software} + \text{Training} + \text{Change Management}Total Costs=Implementation+Software+Training+Change Management

Interpretation:
ROI expresses the steady-state annual financial return after the automation is fully operational. A positive ROI indicates that annualized benefits exceed total costs.

Payback Period (Months)

Payback=Total InvestmentMonthly Net Benefit\text{Payback} = \frac{\text{Total Investment}}{\text{Monthly Net Benefit}}Payback=Monthly Net BenefitTotal Investment Monthly Net Benefit=(Annual Benefits−Recurring Costs)12\text{Monthly Net Benefit} = \frac{(\text{Annual Benefits} - \text{Recurring Costs})}{12}Monthly Net Benefit=12(Annual Benefits−Recurring Costs)

Interpretation:
Measures how many months are required for cumulative net benefits to recover initial

investment, including a Year-1 ramp-up factor to reflect progressive adoption rather than instant realization.

E.2 Net Benefit Formulas

Projection Horizon	Formula	Purpose
Year 1	(Annual Benefits × Ramp-Up Factor) – Year-1 Costs	Reflects partial realization during deployment
3-Year	Σ (Annual Benefits × Year Factor) – Σ (Annual Costs)	Captures cumulative conservative performance
5-Year	Σ (Annual Benefits × Year Factor) – Σ (Annual Costs)	Long-term financial sustainability view

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E.3 FTE and Hours Conversions

FTE Saved=Annual Hours Saved2,080\text{FTE Saved} = \frac{\text{Annual Hours Saved}}{2{,}080}FTE Saved=2,080Annual Hours Saved Annual Hours Saved=(Weekly Process Hours×Employees Involved×Time Savings %)×52\text{Annual Hours Saved} = (\text{Weekly Process Hours} \times \text{Employees Involved} \times \text{Time Savings \%}) \times 52Annual Hours Saved=(Weekly Process Hours×Employees Involved×Time Savings %)×52

Standard Work Year: 2,080 hours (52 weeks × 40 hours)
Interpretation: Converts time saved into full-time-equivalent capacity. Fractional FTEs indicate redeployable time rather than headcount reduction.

E.4 Annual Benefit Breakdown

Total Annual Benefits=Time Savings Benefits+Error Reduction Benefits+Revenue Uplift Benefits\text{Total Annual Benefits} = \text{Time Savings Benefits} + \text{Error Reduction Benefits} + \text{Revenue Uplift Benefits}Total Annual Benefits=Time Savings Benefits+Error Reduction Benefits+Revenue Uplift Benefits

Component	Formula	Description
Time Savings Benefits	Annual Hours Saved × Effective Hourly Rate	Labor cost savings from reduced manual effort

Error Reduction Benefits	(Error Rate Reduction × Cost per Error × Annual Volume) + (Rework Hours Eliminated × Hourly Rate)	Avoided cost of quality failures and rework
Revenue Uplift Benefits	Capacity Hours Reallocated × Revenue per Hour × Conversion Rate	Incremental revenue from redeployed capacity

All benefit calculations apply downward multipliers in conservative modeling to avoid overstatement.

E.5 Scenario Adjustment Framework

To reflect real-world execution realities, Automat-IA applies five adjustment layers that moderate both benefit and cost estimations:

Adjustment Category	Conservative Modeling Rationale
Time Savings Adjustment	Only 70–80 percent of manual effort assumed automatable after considering adoption friction and process exceptions.
Error Reduction Adjustment	Quality improvements capped to account for early-stage variability and partial model coverage.
Revenue Reallocation Adjustment	Assumes a limited portion of freed time converts to revenue-producing work to reflect commercial uncertainty.
Cost Overrun Adjustment	Adds 10–20 percent buffer to account for typical integration and change-management overruns.
Ramp-Up Efficiency Adjustment	Year 1 benefits discounted for gradual adoption; 100 percent run-rate applies from Year 2 onward.

Together these controls form the **Conservative Scenario Framework**, ensuring projections remain achievable, evidence-based, and comparable across diagnostics.

E.6 Key Modeling Assumptions

Variable	Standard Assumption	Rationale

Effective Hourly Rate	1.1 – 1.4 × base salary	Includes benefits, taxes, and overhead
Standard Work Year	2,080 hours	Used for all FTE and productivity conversions
Process Baseline	Validated PIF + interview data	Ensures operational realism
Automation Capability	Derived from TFA readiness levels	Aligns benefits with technical feasibility
Recurring Costs	Constant across projection years	Represents software, licensing, and support
One-Time Costs	Year 1 implementation + training + change management	Captures deployment-phase expenses only

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E.7 Formula Integrity and Audit Trail

All formulas are implemented in Automat-IA's ROI Calculator engine with:

- Version-controlled formula library and revision logs
- Parameter provenance linking each input to respondent and collection date
- Automatic rounding to the nearest hundred for presentation consistency
- Validation checks ensuring all ROI, payback, and FTE values reconcile across appendices

Every figure in this report can be traced back to its originating dataset in **Appendix D: Data Provenance and Audit Trail**.